# ETHAN LEW

https://eth0lew.com

## TECHNICAL SKILLS

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|---|---|
| **Research Experience** | Design Automation, Formal Verification, Cyber-physical Systems |
| **Programming** | Python, Rust, C/C++, CUDA |
| **AI Engineering** | LLM Agents, AI Evaluation |
| **Libraries/Frameworks** | PyTorch, JAX, Gurobi |

## WORK EXPERIENCE

**P-1.ai** June 2025-Present
*Head of Customer Engineering*

· Lead customer engineering, deploying Archie, P-1's AI engineer for the physical world, inside engineering teams' own tools.

**P-1.ai** 2024-June 2025
*Founding Engineer*

· Building Artificial General Engineering Intelligence.

**Galois, Inc., 421 SW 6th Ave, Portland, OR 97204** 2019-2024
*Research Engineer*

· Awarded a Phase I SBIR grant on vehicle autonomy and performed as a principal investigator (PI) on the project.
· Excelled in developing research methodologies, writing sophisticated software prototypes, and exceeding project deliverables.
· Contributed significant research and engineering to high-profile DARPA programs at the company: SSITH (2019-2021), Assured Autonomy (2019-2022), SDCPS (2021-2023), Space-BACN (2022-2023), and FIRE (2024-Present).

**Johns Hopkins APL, 11100 Johns Hopkins Rd, Laurel, MD 20723** 2019
*Electrical Engineering Intern*

· Research Projects: troposcatter communication system feasibility, FDTD solver for bodies of revolution, and methods and benchmarks for an adaptive radar resource manager (RRM).

**Summit Wireless Technologies, 20575 Von Neumann Dr., Beaverton, OR 97006** 2018-2019
*Electrical Engineering Intern*

· Contributed engineering across several teams including the development of a RF power meter, a Linux wireless driver, and the wireless audio stack core.

**Portland State University, 1825 SW Broadway, Portland, OR 97201** 2017
*Climate Research Intern*

· Research Project: *Sensitivity of Global Methane Bayesian Inversion to Surface Observation Data Sets and Chemical-Transport Model Resolution.*
· Presented project at the Center for Climate and Aerosol Research (CCAR) symposium, Council on Undergraduate Research (CUR) symposium, and the American Geophysical Union (AGU) Fall Meeting.

## EDUCATION

**Portland State University, OR** *June 2016 - June 2019*
BS Electrical Engineering *Summa Cum Laude* GPA: 4.00
**Portland Community College, OR** *March 2015 - June 2017*
Transfer Program GPA: 4.00

## AWARDS

**Generalized RAcing Intelligence Competition (GRAIC)** 2022
*1st Place Head-to-Head Category* *CPS-IoT Week*

**Electrical and Computer Engineering Capstone Poster Competition** 2019
*Best Overall Project* *PSU ECE*

**Intel Compute Stick Challenge** 2016
*1st Place* *Intel*

## FUNDED PROJECTS

**SBIR: Phase I: RHEIA-F: Robust High-fidelity Energy-Informed Autonomy Framework** 2024
*PI: **Ethan Lew**; Co-PI: Nicola Bezzo* *AFRL Funded, Galois Inc. Award: $179,934*

· We propose the Robust High-fidelity Energy-Informed Autonomy Framework (RHEIA-F), an advanced energy-aware mission planning framework for unmanned aerial systems (UAS). This framework integrates comprehensive energy management into UAS missions, providing both off-board and on-board components. These capabilities are directly transferable to other Department of Defense (DoD) embedded systems, autonomous vehicles, and space systems.

## PROGRAM COMMITTEES

**26th ACM International Conference on Hybrid Systems: Computation and Control** 2021
*Repeatability Evaluation Program Committee* *HSCC 2023*

**25th ACM International Conference on Hybrid Systems: Computation and Control** 2021
*Repeatability Evaluation Program Committee* *HSCC 2022*

**7th IFAC Conference on Analysis and Design of Hybrid Systems** 2021
*Repeatability Evaluation Program Committee* *ADHS 2021*

**24th ACM International Conference on Hybrid Systems: Computation and Control** 2021
*Repeatability Evaluation Program Committee* *HSCC 2021*

## PEER-REVIEWED PUBLICATIONS

**(authors alphabetical)** Bak, S., Bogomolov, S., Hencey, B., Kochdumper, N., **Lew, E.**, and Potomkin, K. (2025a). Reachability of koopman linearized systems using explicit kernel approximation and polynomial zonotope refinement. *Formal Methods in System Design*, pages 1–27

**(authors alphabetical)** Bak, S., Hekal, A., Kochdumper, N., **Lew, Ethan**, Mata, A., and Rahmati, A. (2025b). Fast koopman surrogate falsification using linear relaxations and weights. In Akshay, S., Niemetz, A., and Sankaranarayanan, S., editors, *Automated Technology for Verification and Analysis*, pages 234–255, Cham. Springer Nature Switzerland

Khandait, T., Formica, F., Arcaini, P., Chotaliya, S., Fainekos, G., Hekal, A., Kundu, A., **Lew, E.**, Loreti, M., Menghi, C., et al. (2024). Arch-comp 2024 category report: Falsification. In *Proceedings of the 11th Int. Workshop on Applied*, volume 103, pages 122–144

**(authors alphabetical)** Abowd, J. M., Adams, T., Ashmead, R., Darais, D., Dey, S., Garfinkel, S. L., Goldschlag, N., Kifer, D., Leclerc, P., **Lew, E.**, et al. (2023). The 2010 census confidentiality protections failed, here's how and why. Technical report, National Bureau of Economic Research

**(authors alphabetical)** Bak, S., Bogomolov, S., Hekal, A., Kochdumper, N., **Lew, E.**, Mata, A., and Rahmati, A. (2024). Falsification using reachability of surrogate koopman models. In *Proceedings of the 27th ACM International Conference on Hybrid Systems: Computation and Control*, pages 1–13

Lahouel, K., Wells, M., Rielly, V., **Lew, E.**, Lovitz, D., and Jedynak, B. M. (2024). Learning non-parametric ordinary differential equations from noisy data. *Journal of Computational Physics*, page 112971

**Lew, E.**, Hekal, A., Potomkin, K., Kochdumper, N., Hencey, B., Bak, S., and Bogomolov, S. (2023). Autokoopman: A toolbox for automated system identification via koopman operator linearization. In André, É. and Sun, J., editors, *Automated Technology for Verification and Analysis*, pages 237–250, Cham. Springer Nature Switzerland

Davis, E., Dey, S., Karvonen, A., **Lew, E.**, Quick, D., Shyamshankar, P., Hille, T., and Lebeau, M. (2023). Leveraging manifold learning and relationship equity management for symbiotic explainable artificial intelligence. In *International Conference on Applied Human Factors and Ergonomics*, pages 490–510. AHFE International

**(authors alphabetical)** Bak, S., Bogomolov, S., Hencey, B., Kochdumper, N., **Lew, E.**, and Potomkin, K. (2022). Reachability of koopman linearized systems using random fourier feature observables and polynomial zonotope refinement. In *International Conference on Computer Aided Verification*, pages 490–510. Springer